Hidden Markov Models

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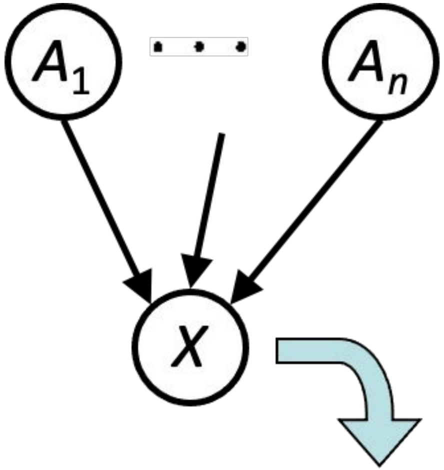
Some of Slides have been adopted from Berkeley CS188 & Sharif UT CE417

Bayes Net

- A directed acyclic graph (DAG) with a set of nodes one per variable
- Each link from X to Y shows a "direct influence" of X on Y
- For each node, a conditional probability distribution **$P(X_i|Parents(X_i))$** by CPT (conditional probability table)



$$P(X|A_1 \ldots A_n)$$

A Bayes net = Topology (graph) + Local Conditional Probabilities

Two Equivalent Assumptions of Bayes Nets

- Factorization Property:

$$P(x_1, x_2, \ldots x_n) = \prod_{i=1}^{n} P(x_i | \mathit{parents}(X_i))$$

- Local Markov Property:

$$X_i \perp \text{NonDescendants}(X_i) \mid \text{Parents}(X_i)$$

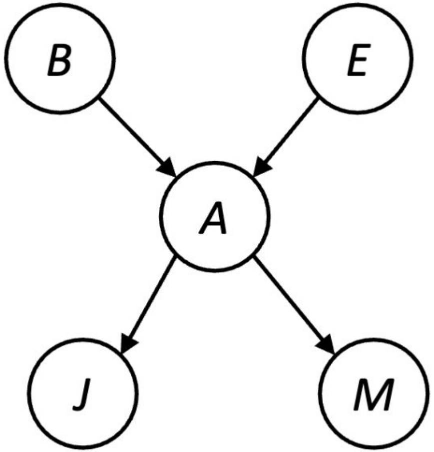
Two Types of Queries

- Asking about Conditional Independencies: Is X conditionally independent of Y give {Z}?
    - Solution: D-Separation Algorithm

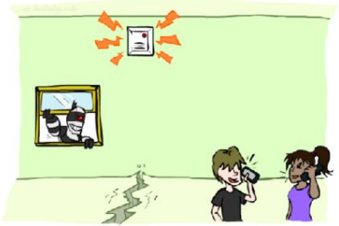
- Probabilistic Inference: $P(Q|E_1 = e_1, \ldots E_k = e_k)$
    - Solutions: Inference By Enumeration and Variable Elimination

Example: Alarm Network

B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998



A	J	P(J|A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M|A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

B	E	A	P(A|B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

$$P(+b, -e, +a, -j, +m) =$$
$$P(+b)P(-e)P(+a|+b,-e)P(-j|+a)P(+m|+a) =$$
$$0.001 \times 0.998 \times 0.94 \times 0.1 \times 0.7$$

Sampling From a Distribution

C	P(C)
red	0.6
green	0.1
blue	0.3

Step 1: Get sample u from uniform distribution over [0, 1)

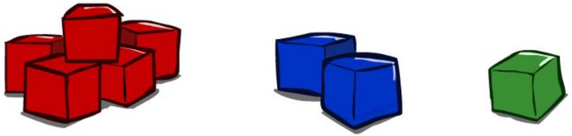
- E.g. random() in python

Step 2:

$$\begin{aligned} 0 \le u < 0.6, &\rightarrow C = red \\ 0.6 \le u < 0.7, &\rightarrow C = green \\ 0.7 \le u < 1, &\rightarrow C = blue \end{aligned}$$

If random() returns u=0.83, then our sample is C=blue

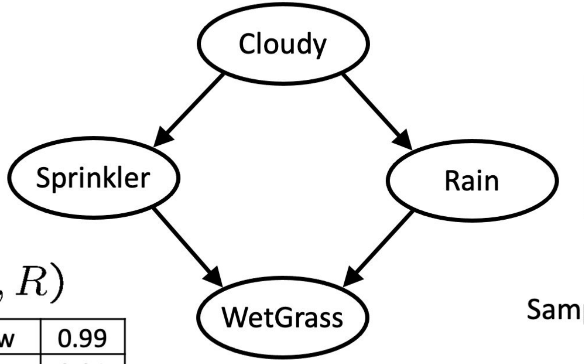
E.g, after sampling 8 times:



Sampling From a Bayes Net: Prior Sampling

$P(C)$

+c	0.5
-c	0.5



$P(S|C)$

+c	+s	0.1
	-s	0.9
-c	+s	0.5
	-s	0.5

$P(R|C)$

+c	+r	0.8
	-r	0.2
-c	+r	0.2
	-r	0.8

$P(W|S,R)$

+s	+r	+w	0.99
		-w	0.01
	-r	+w	0.90
		-w	0.10
-s	+r	+w	0.90
		-w	0.10
	-r	+w	0.01
		-w	0.99

Samples:

Sampling From a Bayes Net: Prior Sampling (cont.)

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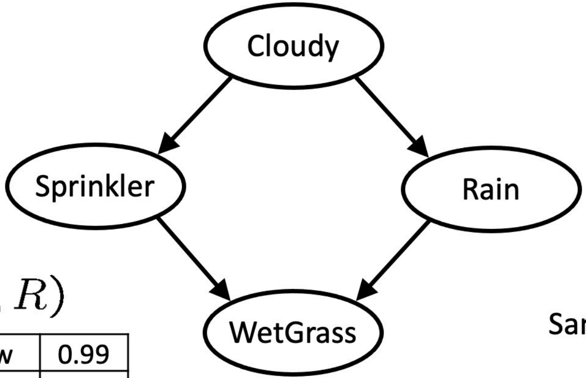

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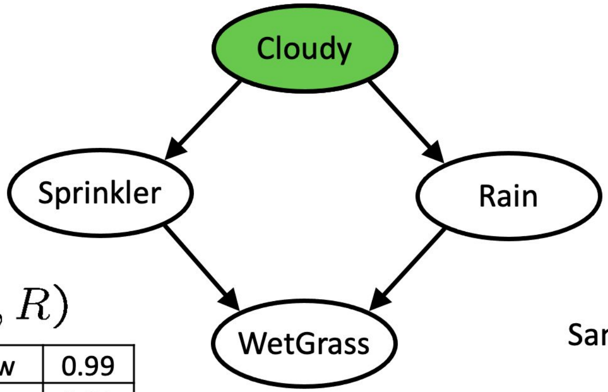

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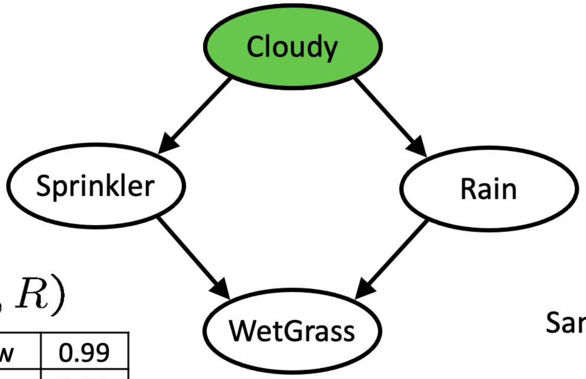

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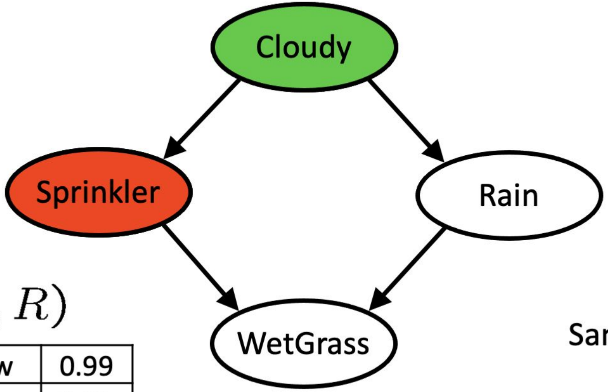

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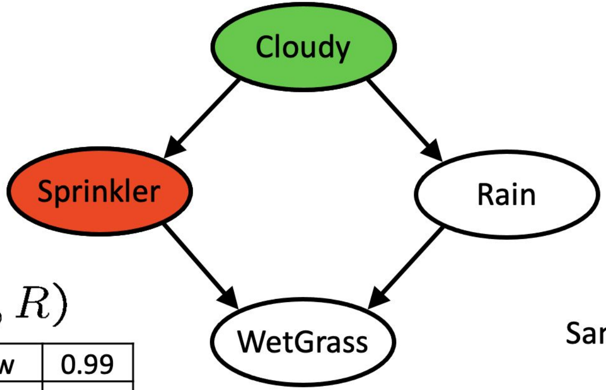

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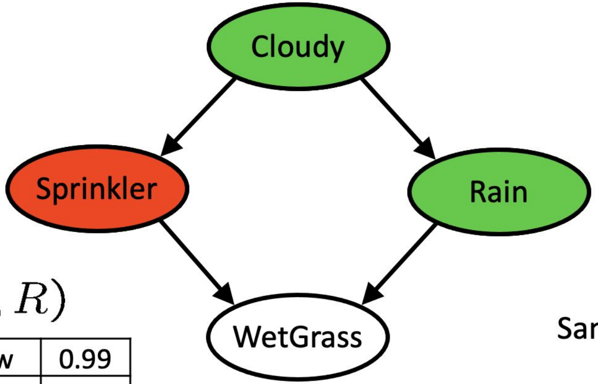

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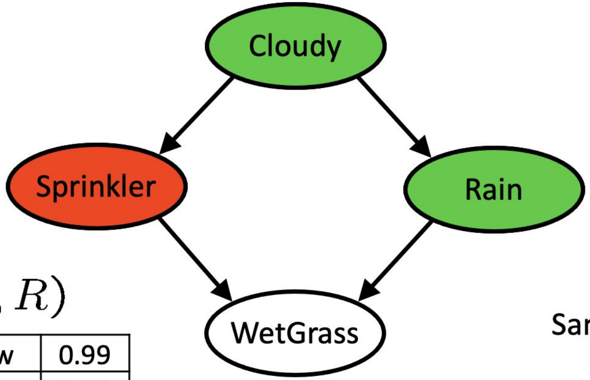

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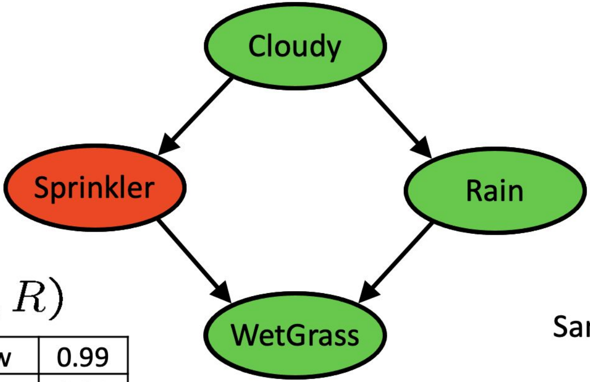

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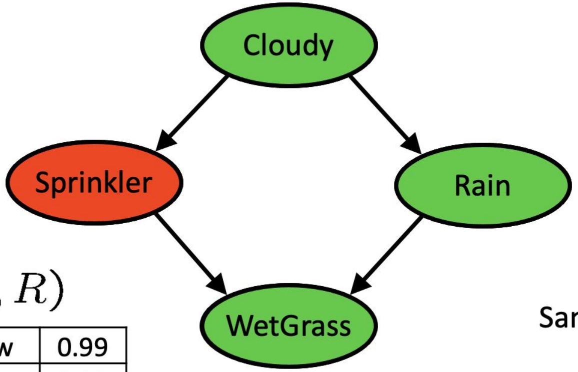

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Samples:

+c, -s, +r, +w

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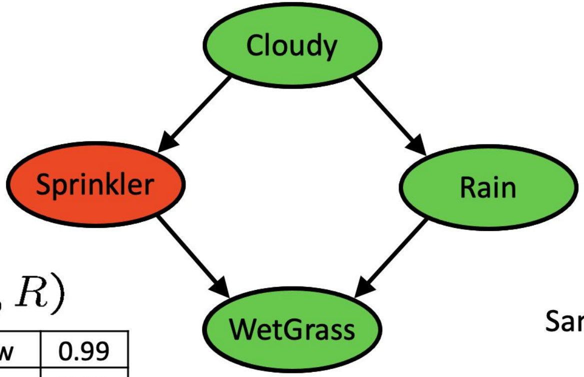

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Samples:

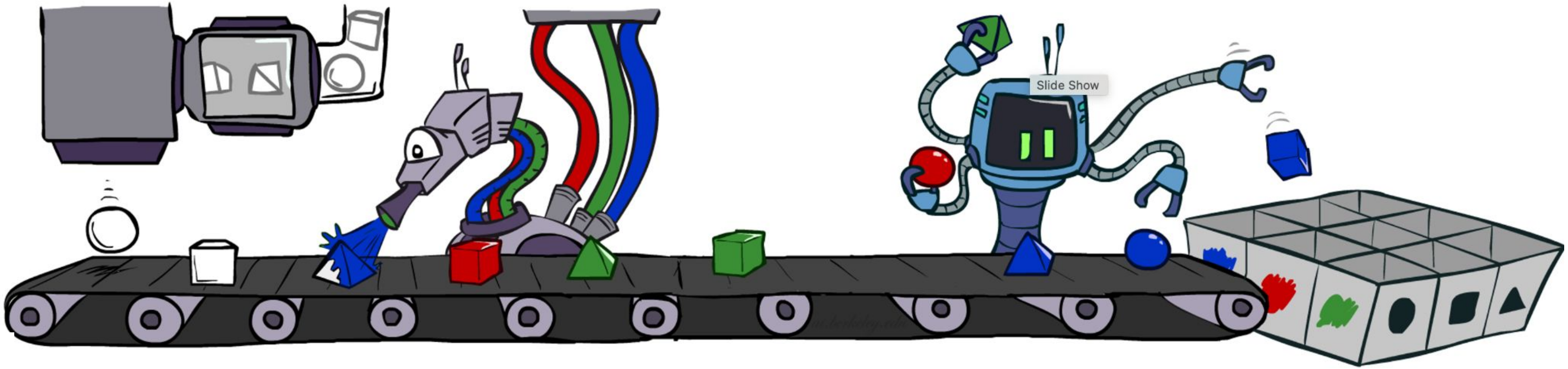
+c, -s, +r, +w

-c, +s, -r, +w

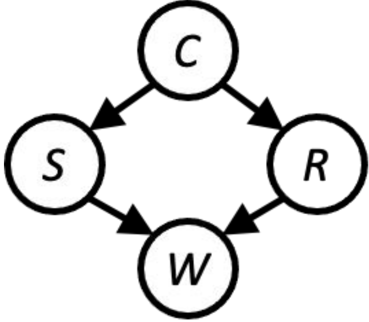
...

Prior Sampling

- For $i=1, 2, \ldots, n$
  - Sample x_i from $P(X_i \mid Parents(X_i))$
- Return $(x_1, x_2, \ldots, x_n)$



Usage of Sampling



We'll get a bunch of samples from the BN:

+c, -s, +r, +w

+c, +s, +r, +w

-c, +s, +r, -w

+c, -s, +r, +w

-c, -s, -r, +w

If we want to know P(W)
We have counts <+w:4, -w:1>

Normalize to get P(W) = <+w:0.8, -w:0.2>

This will get closer to the true distribution with more samples

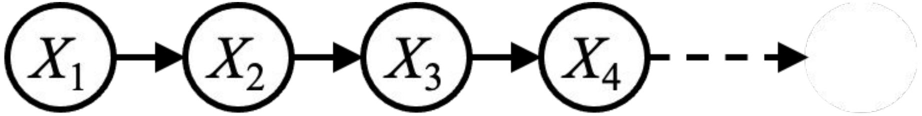
Can estimate anything else, too
What about P(C| +w)? P(C| +r, +w)? P(C| -r, -w)?
Fast: getting a sample is faster than computing the right answer (e.g. with variable elimination)

Reasoning over Time or Space

- Often, we want to reason about a sequence of observations:
  - Speech recognition
  - Robot localization
  - User attention
  - Medical monitoring
- Need to introduce time (or space) into our models

Markov Chain

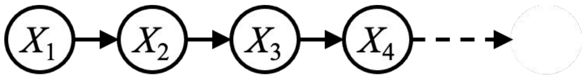
- Value of X at a given time is called the state



$P(X_1)$ $P(X_t|X_{t-1})$

- Parameters: called **transition probabilities** or dynamics, specify how the state evolves over time (also, **initial state probabilities**)
- Stationarity assumption: transition probabilities the same at all times

Joint Distribution of a Markov Chain



$P(X_1)$ $\quad$ $P(X_t|X_{t-1})$

- Joint distribution:

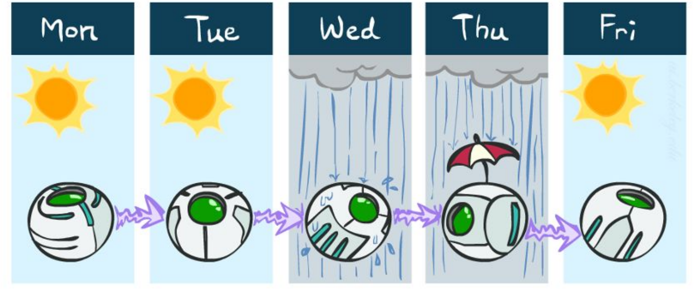
$$P(X_1, X_2, \ldots, X_T) = P(X_1)P(X_2|X_1)P(X_3|X_2)\ldots P(X_T|X_{T-1})$$
$$= P(X_1)\prod_{t=2}^{T} P(X_t|X_{t-1})$$

Conditional Independence in Markov Chain

- Basic conditional independence:
  - Past and future independent of the present
  - Each time step only depends on the previous
  - This is called the **(first order) Markov property**
- Note that the chain is just a (growable) BN
- We can always use generic BN reasoning on it if we truncate the chain at a fixed length

Example Markov Chain: Weather

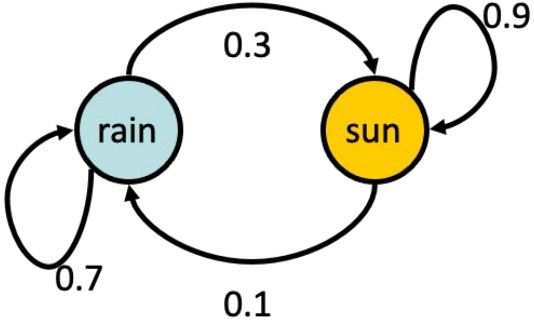
- States: X = {rain, sun}

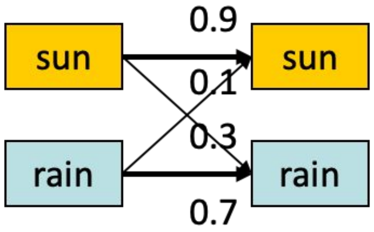


- Initial distribution (X_1): 1.0 sun
- CPT $P(X_t \mid X_{t-1})$:

Two new ways of representing the same CPT

X_{t-1}	X_t	$P(X_t \mid X_{t-1})$
sun	sun	0.9
sun	rain	0.1
rain	sun	0.3
rain	rain	0.7



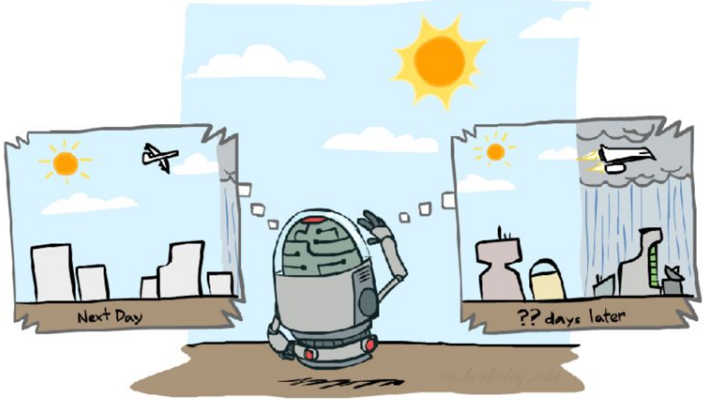


Inference Example

- Question: What's P(X) on some day t?

$$X_1 \rightarrow X_2 \rightarrow X_3 \rightarrow X_4 \dashrightarrow \quad X_t \ ?$$

- We know $P(X_1)$ and $P(X_t|X_{t-1})$



$$P(X_1) = \text{known}$$

$$P(x_t) = \sum_{x_{t-1}} P(x_{t-1}, x_t)$$
$$= \sum_{x_{t-1}} P(x_t \mid x_{t-1}) P(x_{t-1})$$

Forward simulation

Inference Example

- Initial distribution: 1.0 sun
  - We know: $P(X_1)$ $P(X_t|X_{t-1})$

X_{t-1}	X_t	$P(X_t\|X_{t-1})$
sun	sun	0.9
sun	rain	0.1
rain	sun	0.3
rain	rain	0.7

- What is the probability distribution after one step?

$$P(X_2 = sun) = \sum_{x_1} P(x_1, X_2 = sun) = \sum_{x_1} P(X_2 = sun|x_1)P(x_1)$$

$$= \quad P(X_2 = \text{sun}|X_1 = \text{sun})P(X_1 = \text{sun}) + P(X_2 = \text{sun}|X_1 = \text{rain})P(X_1 = \text{rain})$$

$$0.9 \cdot 1.0 + 0.3 \cdot 0.0 = 0.9$$

Example Run of Mini-Forward Algorithm

From initial observation of sun

$$\underset{P(X_1)}{\begin{pmatrix} 1.0 \\ 0.0 \end{pmatrix}} \quad \underset{P(X_2)}{\begin{pmatrix} 0.9 \\ 0.1 \end{pmatrix}} \quad \underset{P(X_3)}{\begin{pmatrix} 0.84 \\ 0.16 \end{pmatrix}} \quad \underset{P(X_4)}{\begin{pmatrix} 0.804 \\ 0.196 \end{pmatrix}} \quad \rightarrow \quad \underset{P(X_\infty)}{\begin{pmatrix} 0.75 \\ 0.25 \end{pmatrix}}$$

From initial observation of rain

$$\underset{P(X_1)}{\begin{pmatrix} 0.0 \\ 1.0 \end{pmatrix}} \quad \underset{P(X_2)}{\begin{pmatrix} 0.3 \\ 0.7 \end{pmatrix}} \quad \underset{P(X_3)}{\begin{pmatrix} 0.48 \\ 0.52 \end{pmatrix}} \quad \underset{P(X_4)}{\begin{pmatrix} 0.588 \\ 0.412 \end{pmatrix}} \quad \rightarrow \quad \underset{P(X_\infty)}{\begin{pmatrix} 0.75 \\ 0.25 \end{pmatrix}}$$

From yet another initial distribution $P(X_1)$:

$$\underset{P(X_1)}{\begin{pmatrix} p \\ 1-p \end{pmatrix}} \quad \ldots \quad \rightarrow \quad \underset{P(X_\infty)}{\begin{pmatrix} 0.75 \\ 0.25 \end{pmatrix}}$$

Stationary Distributions

For most chains:

Influence of the initial distribution gets less and less over time.

The distribution we end up in is independent of the initial distribution.

Stationary distribution:

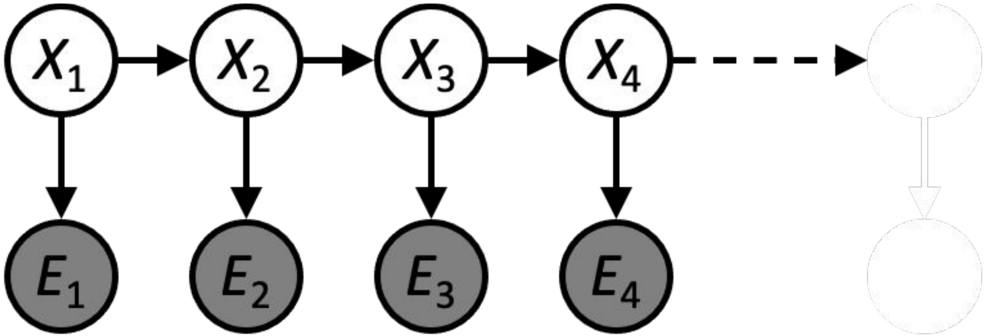
The distribution we end up with is called the stationary distribution P_∞ of the chain

It satisfies:

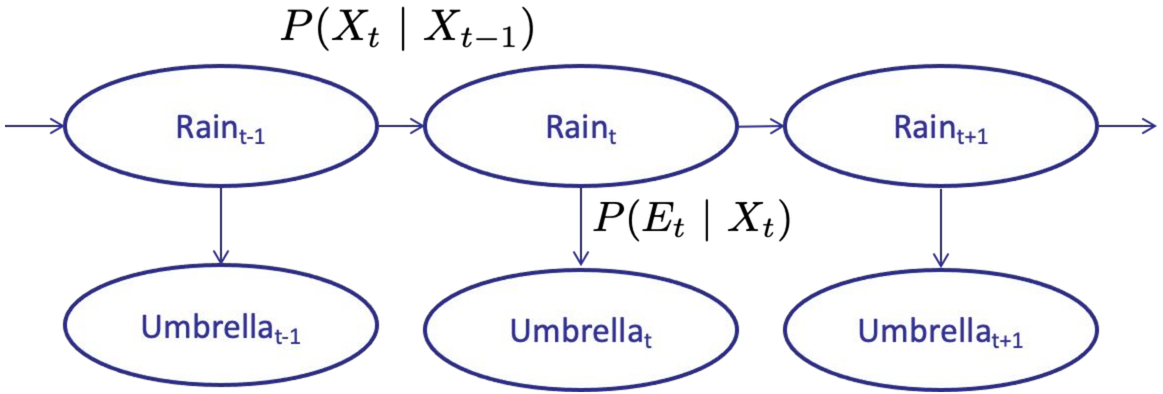
$$P_\infty(X) = P_{\infty+1}(X) = \sum_x P(X|x) P_\infty(x)$$

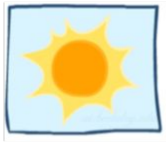
Hidden Markov Models (HMM)

- Underlying Markov chain over states X
- You observe outputs (effects) at each time step



Example: Weather HMM





An HMM is defined by:

Initial distribution: $P(X_1)$

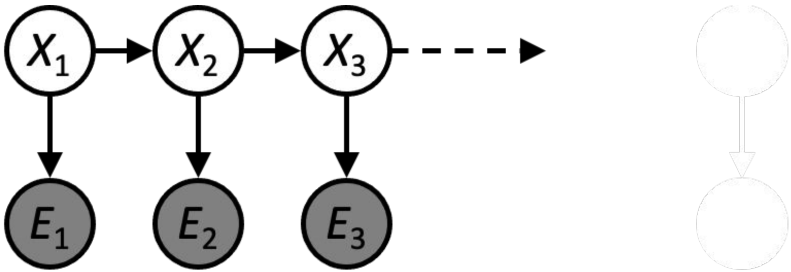
Transitions: $P(X_t \mid X_{t-1})$

Emissions: $P(E_t \mid X_t)$

R_t	R_{t+1}	$P(R_{t+1} \mid R_t)$
+r	+r	0.7
+r	-r	0.3
-r	+r	0.3
-r	-r	0.7

R_t	U_t	$P(U_t \mid R_t)$
+r	+u	0.9
+r	-u	0.1
-r	+u	0.2
-r	-u	0.8

Joint Distribution of an HMM

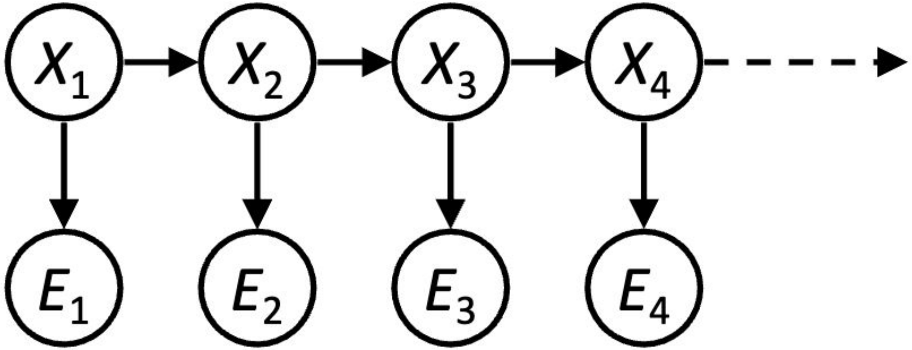


- Joint distribution:

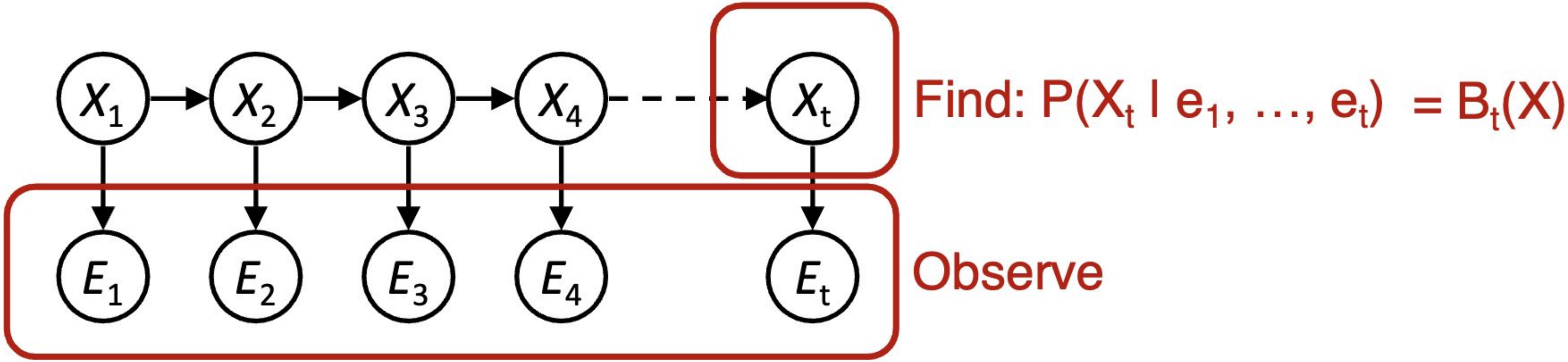
$$P(X_1, E_1, \ldots, X_T, E_T) = P(X_1)P(E_1|X_1) \prod_{t=2}^{T} P(X_t|X_{t-1})P(E_t|X_t)$$

Conditional Independence in HMMs

- Future states are independent of past states, given current state
- Current observation is independent of all else, given current state

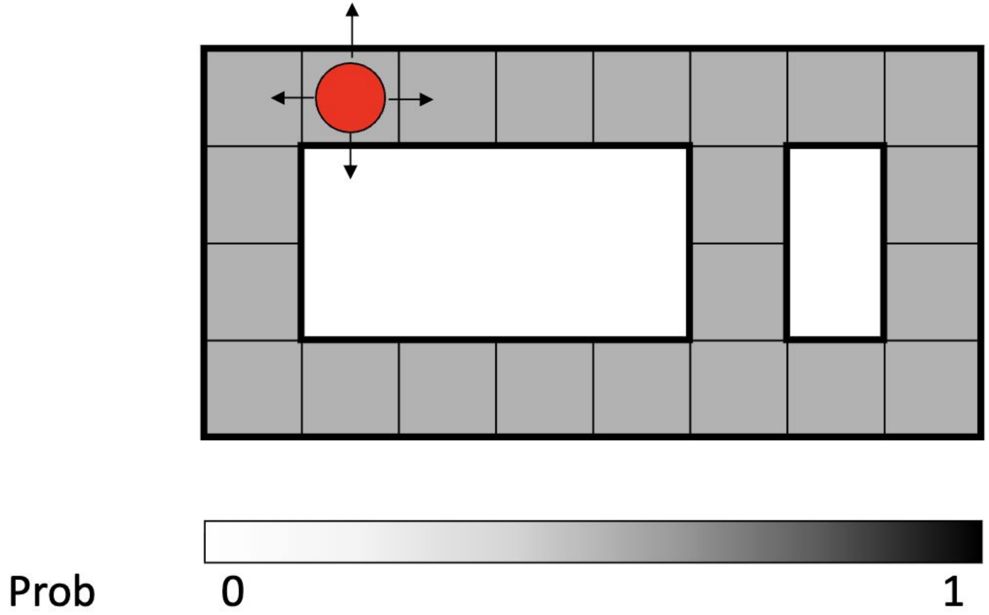


Inference in HMMs: Filtering



- Filtering, or monitoring, is the task of tracking the distribution $B_t(X) = P_t(X_t \mid e_1, \ldots, e_t)$ (the belief state) over time
- We start with $B_1(X)$ in an initial setting, usually uniform
  - As time passes, or we get observations, we update $B(X)$

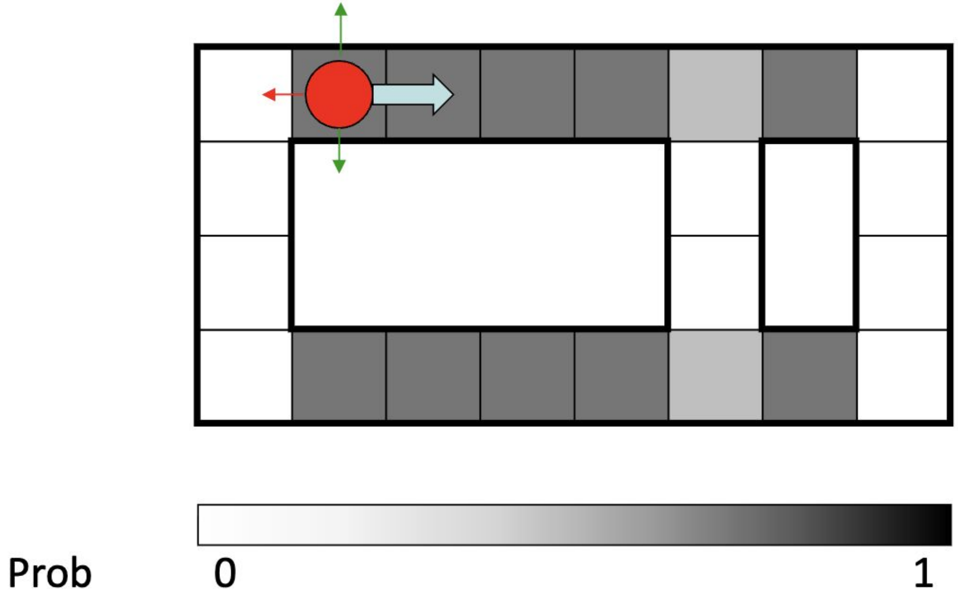
Example: Robot Localization



t=0

Sensor model: can read in which directions there is a wall, never more than 1 mistake

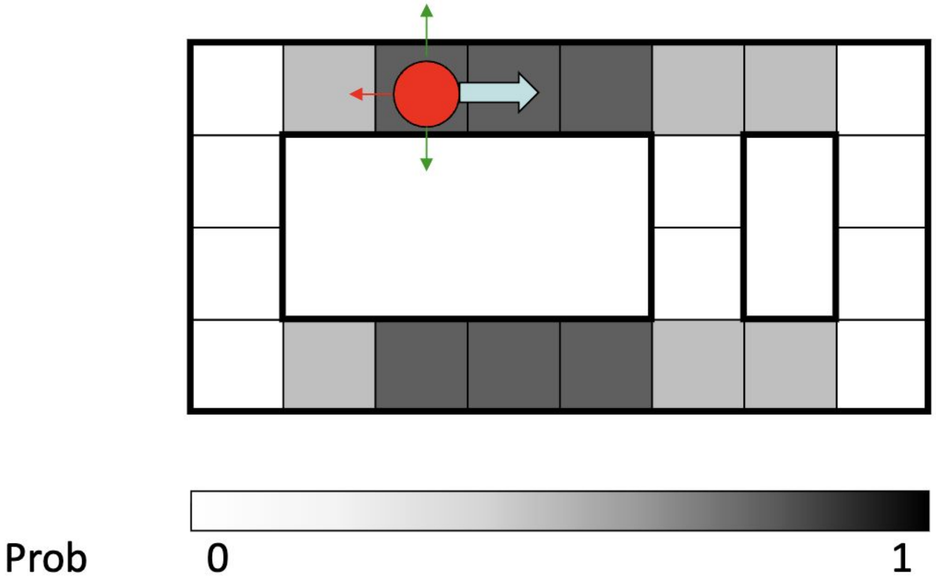
Example: Robot Localization (cont.)



t=1

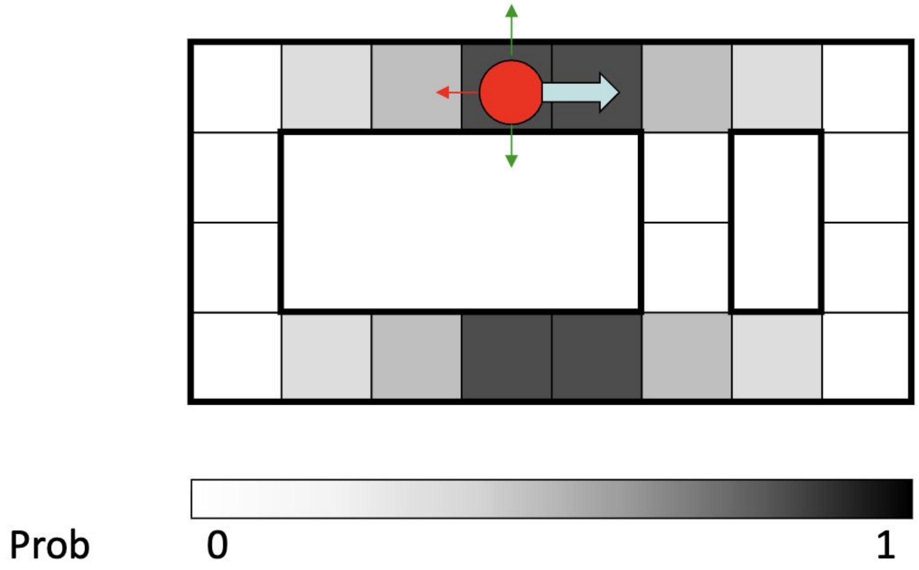
Lighter grey: was possible to get the reading, but less likely b/c required 1 mistake

Example: Robot Localization (cont.)



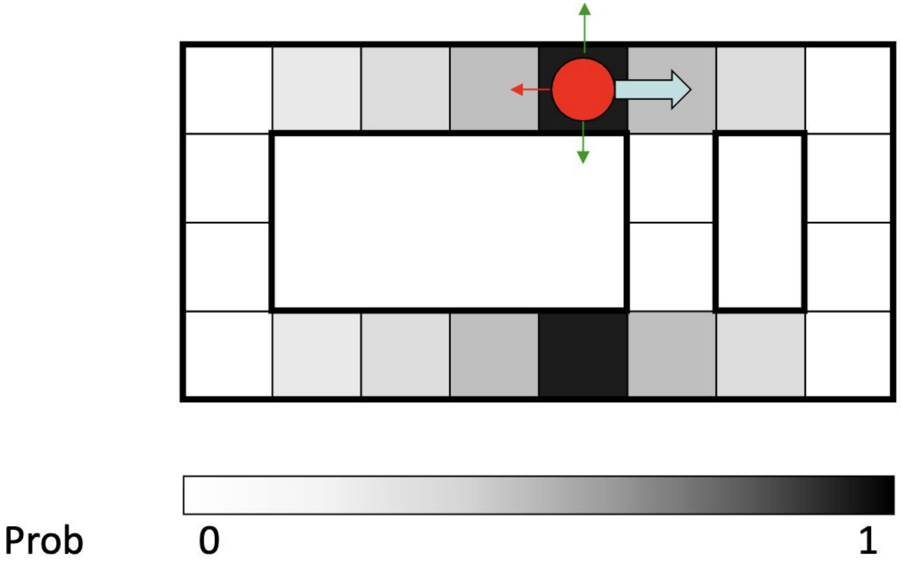
t=2

Example: Robot Localization (cont.)



t=3

Example: Robot Localization (cont.)



t=4

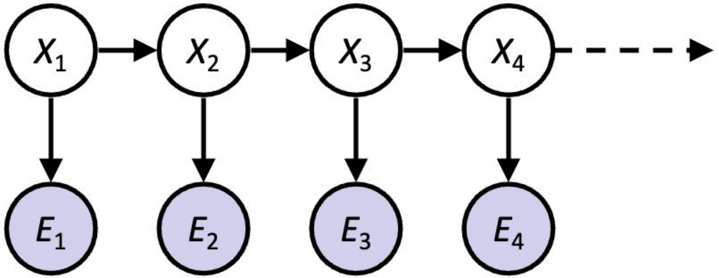
HMM Inference: Find State Given Evidence

- We are given evidence at each time and want to know

$$B_t(X) = P(X_t | e_{1:t})$$

- Idea: start with $P(X_1)$ and derive $B_t(X)$ in terms of $B_{t-1}(X)$
  - Two steps: **Passage of Time** & **Observation**

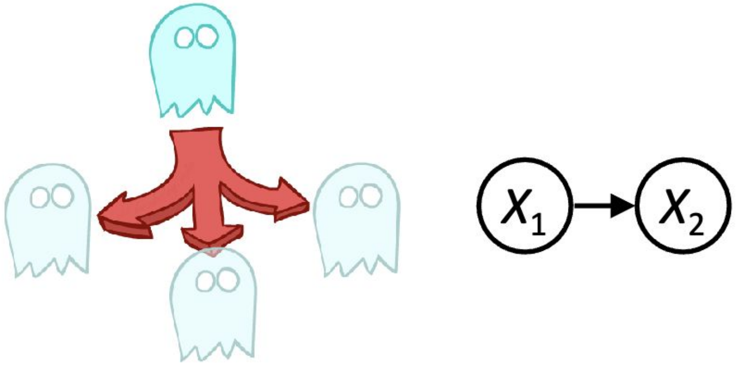
$B'_4(X) = P(X_4 | e_{1:3})$



$B_3(X)$ $\quad$ $B_4(X) = P(X_4 | e_{1:4})$

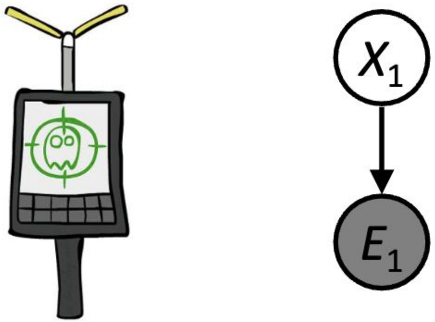
Two Steps in HMM Inference

Passage of Time:



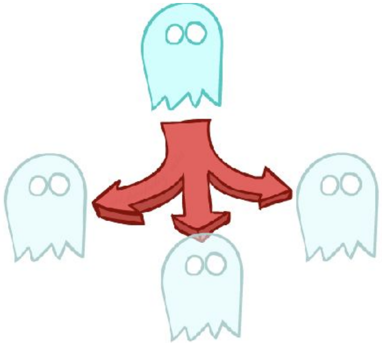
$P(X_2)$

Observation:



$P(X_1|e_1)$

Passage of Time: Base Case



$$X_1 \rightarrow X_2$$

Have: $P(X_1)$ $\quad$ $P(X_2|X_1)$

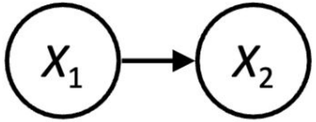
Want: $P(X_2)$

$$\begin{aligned} P(x_2) &= \sum_{x_1} P(x_1, x_2) \\ &= \sum_{x_1} P(x_1)P(x_2|x_1) \end{aligned}$$

Passage of Time: General Case

- Assume we have current belief P(X | evidence to date)

$$B(X_t) = P(X_t|e_{1:t})$$



- Then, after one time step passes:

$$P(X_{t+1}|e_{1:t}) = \sum_{x_t} P(X_{t+1}, x_t|e_{1:t})$$

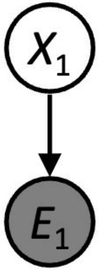
$$= \sum_{x_t} P(X_{t+1}|x_t, e_{1:t})P(x_t|e_{1:t})$$

$$= \sum_{x_t} P(X_{t+1}|x_t)P(x_t|e_{1:t})$$

- Or compactly:

$$B'(X_{t+1}) = \sum_{x_t} P(X'|x_t)B(x_t)$$

Observation: Base Case



Have: $P(X_1)$ $P(E_1|X_1)$

Want: $P(X_1|e_1)$

$$P(x_1|e_1) = P(x_1, e_1)/P(e_1)$$
$$\propto_{X_1} P(x_1, e_1)$$
$$= P(x_1)P(e_1|x_1)$$

Also can write as:

$$P(x_1|e_1) = \frac{P(x_1)P(e_1|x_1)}{\sum_{x'} P(x')P(e_1|x')}$$

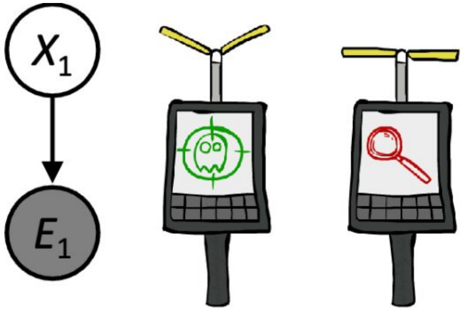
Observation: General Case

- Assume we have current belief P(X | previous evidence):

$$B'(X_{t+1}) = P(X_{t+1}|e_{1:t})$$

- Then, after evidence comes in:

$$\begin{aligned} P(X_{t+1}|e_{1:t+1}) &= P(X_{t+1}, e_{t+1}|e_{1:t})/P(e_{t+1}|e_{1:t}) \\ &\propto_{X_{t+1}} P(X_{t+1}, e_{t+1}|e_{1:t}) \\ &= P(e_{t+1}|e_{1:t}, X_{t+1})P(X_{t+1}|e_{1:t}) \\ &= P(e_{t+1}|X_{t+1})P(X_{t+1}|e_{1:t}) \end{aligned}$$



- Or, compactly:

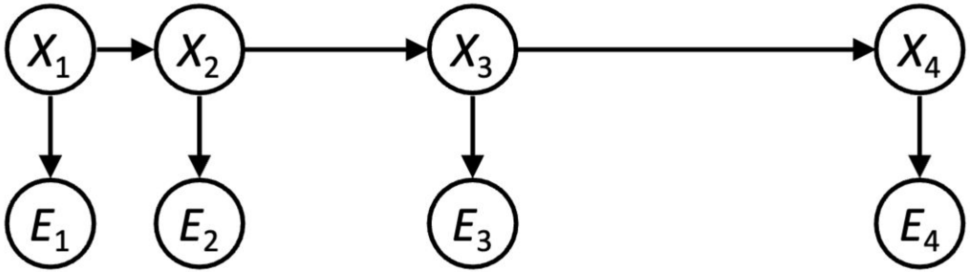
$$B(X_{t+1}) \propto_{X_{t+1}} P(e_{t+1}|X_{t+1})B'(X_{t+1})$$

Two Steps: Passage of Time + Observation

Evidence at each time: +u

$$B(X_t) = P(X_t|e_{1:t})$$

$$B'(X_{t+1}) = \sum_{x_t} P(X'|x_t)B(x_t)$$



$$B(X_{t+1}) \propto_{X_{t+1}} P(e_{t+1}|X_{t+1})B'(X_{t+1})$$

Example: Weather HMM Inference

Evidence at each time: +u

B'(+r) = ?
B'(-r) = ?

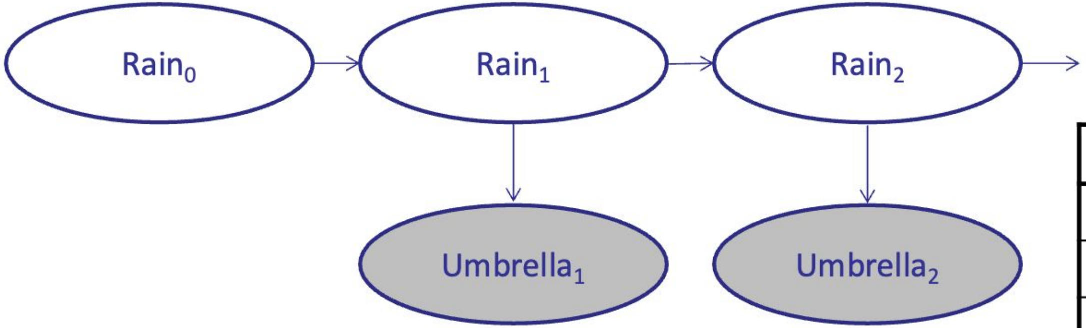
B(+r) = 0.5
B(-r) = 0.5

Passage of Time:

$$B'(X_{t+1}) = \sum_{x_t} P(X_{t+1}|x_t)B(x_t)$$

Observation:

$$B(X_{t+1}) \propto_{X_{t+1}} P(e_{t+1}|X_{t+1})B'(X_{t+1})$$



$P(X_{t+1}|X_t)$

R_t	R_{t+1}	$P(R_{t+1}\|R_t)$
+r	+r	0.7
+r	-r	0.3
-r	+r	0.3
-r	-r	0.7

$P(E_t|X_t)$

R_t	U_t	$P(U_t\|R_t)$
+r	+u	0.9
+r	-u	0.1
-r	+u	0.2
-r	-u	0.8

Example: Weather HMM Inference


Evidence at each time: +u

B'(+r) = 0.5*0.7 + 0.5*0.3 = 0.5
B'(-r) = 0.5*0.3 + 0.5*0.7 = 0.5

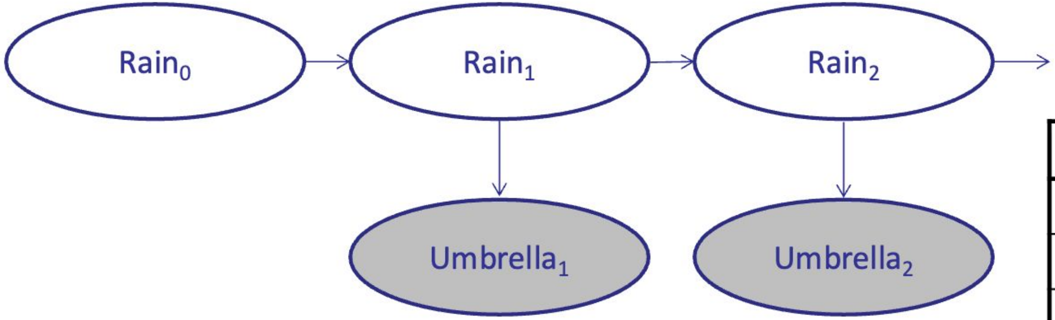
B(+r) = 0.5
B(-r) = 0.5

Passage of Time:

$$B'(X_{t+1}) = \sum_{x_t} P(X_{t+1}|x_t)B(x_t)$$

Observation:

$$B(X_{t+1}) \propto_{X_{t+1}} P(e_{t+1}|X_{t+1})B'(X_{t+1})$$



$P(X_{t+1}|X_t)$

R_t	R_{t+1}	$P(R_{t+1}\|R_t)$
+r	+r	0.7
+r	-r	0.3
-r	+r	0.3
-r	-r	0.7

$P(E_t|X_t)$

R_t	U_t	$P(U_t\|R_t)$
+r	+u	0.9
+r	-u	0.1
-r	+u	0.2
-r	-u	0.8

Example: Weather HMM Inference

Evidence at each time: +u

B(+r) = 0.5
B(-r) = 0.5

B'(+r) = 0.5
B'(-r) = 0.5

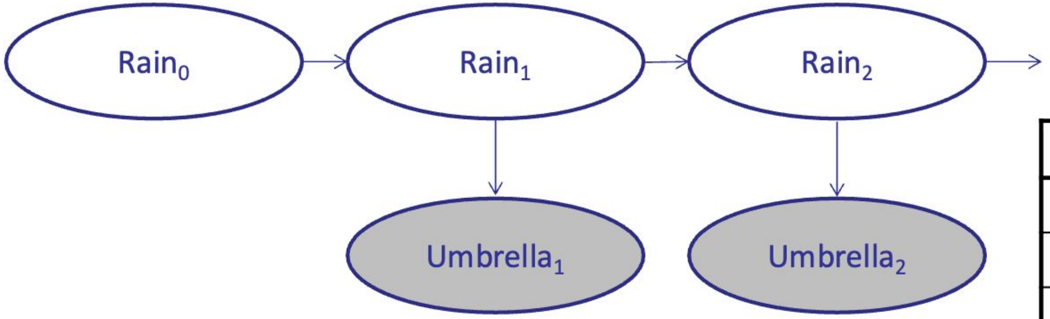
B(+r) = ?
B(-r) = ?

Passage of Time:

$$B'(X_{t+1}) = \sum_{x_t} P(X_{t+1}|x_t)B(x_t)$$

Observation:

$$B(X_{t+1}) \propto_{X_{t+1}} P(e_{t+1}|X_{t+1})B'(X_{t+1})$$



$P(X_{t+1}|X_t)$

R_t	R_{t+1}	$P(R_{t+1}\|R_t)$
+r	+r	0.7
+r	-r	0.3
-r	+r	0.3
-r	-r	0.7

$P(E_t|X_t)$

R_t	U_t	$P(U_t\|R_t)$
+r	+u	0.9
+r	-u	0.1
-r	+u	0.2
-r	-u	0.8

Example: Weather HMM Inference

Evidence at each time: +u

Passage of Time:

$$B'(X_{t+1}) = \sum_{x_t} P(X_{t+1}|x_t) B(x_t)$$

Observation:

$$B(X_{t+1}) \propto_{X_{t+1}} P(e_{t+1}|X_{t+1}) B'(X_{t+1})$$

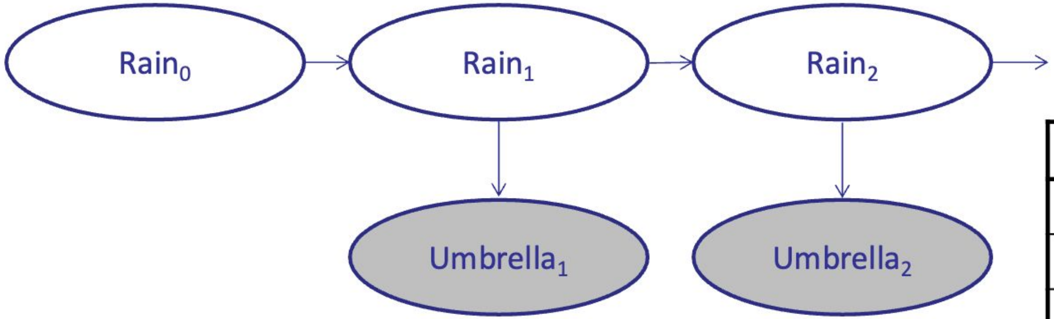
B(+r) = 0.5
B(-r) = 0.5

B'(+r) = 0.5
B'(-r) = 0.5

B(+r) = 0.9*0.5 = 0.45
B(-r) = 0.2*0.5 = 0.10

normalize

0.818
0.182



$P(X_{t+1}|X_t)$

R_t	R_{t+1}	$P(R_{t+1}\|R_t)$
+r	+r	0.7
+r	-r	0.3
-r	+r	0.3
-r	-r	0.7

$P(E_t|X_t)$

R_t	U_t	$P(U_t\|R_t)$
+r	+u	0.9
+r	-u	0.1
-r	+u	0.2
-r	-u	0.8

Example: Weather HMM Inference

Evidence at each time: +u

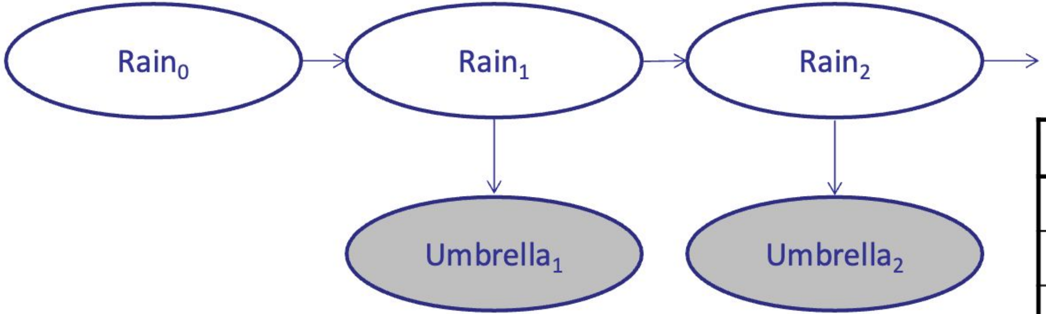
B(+r) = 0.5
B(-r) = 0.5

B'(+r) = 0.5
B'(-r) = 0.5

B(+r) = 0.818
B(-r) = 0.182

B'(+r) = 0.627
B'(-r) = 0.373

B(+r) = 0.883
B(-r) = 0.117



Passage of Time:

$$B'(X_{t+1}) = \sum_{x_t} P(X_{t+1}|x_t)B(x_t)$$

Observation:

$$B(X_{t+1}) \propto_{X_{t+1}} P(e_{t+1}|X_{t+1})B'(X_{t+1})$$

$P(X_{t+1}|X_t)$

R_t	R_{t+1}	$P(R_{t+1}\|R_t)$
+r	+r	0.7
+r	-r	0.3
-r	+r	0.3
-r	-r	0.7

$P(E_t|X_t)$

R_t	U_t	$P(U_t\|R_t)$
+r	+u	0.9
+r	-u	0.1
-r	+u	0.2
-r	-u	0.8

Forward Algorithm: Combining Two Steps

Elapse time: compute $P(X_t \mid e_{1:t-1})$

$$P(x_t|e_{1:t-1}) = \sum_{x_{t-1}} P(x_{t-1}|e_{1:t-1}) \cdot P(x_t|x_{t-1})$$

Observe: compute $P(X_t \mid e_{1:t})$

$$P(x_t|e_{1:t}) \propto P(x_t|e_{1:t-1}) \cdot P(e_t|x_t)$$

Next Session: Learning – Naive Bayes